Arista Access Node MG-ACC-38 Routing Protocol Flap SOP

Node: MG-ACC-38 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_9198

Description

This Standard Operating Procedure (SOP) addresses the critical failure domain of Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on the Arista Access Node MG-ACC-38, situated at the MG Road Core site. The root cause of this failure domain lies in the complex interplay between the Routing Information Base (RIB) and the Forwarding Information Base (FIB) within the Arista EOS operating system. When the BGP route churn, OSPF adjacency loss, or ISIS LSP flooding occurs, the RIB attempts to converge, leading to a temporary state of uncertainty, which in turn causes the FIB to recompute the forwarding table. This process, although designed to ensure network stability, can cause a routing protocol flap, resulting in packet loss, increased latency, and potential network downtime. The Arista Access Node, being a critical juncture in the network topology, exacerbates this issue due to its role in aggregating and distributing traffic to and from the core network.

Pre-requisite Checks

- Ensure remote hands are dispatched to the MG Road Core site.

- Verify the status of redundant links and confirm they are operational.

- Check the power supply and cooling systems for the Arista Access Node to ensure they are functioning within normal parameters.

- Review the network topology and identify any recent changes that may have contributed to the routing protocol flap.

Potential Root Causes

Cause	Probability
BGP Route Churn	High
OSPF Adjacency Loss	Medium
ISIS LSP Flooding	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP table version is 2, main routing table version is 2 Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State | bgp.200.0.0.1 4 200 10 10 2 0 0 00:02:11 4w4d 4 10/10, Description: This command displays a summary of the BGP neighbor connections, including the number of messages received and sent, as well as the table version. By analyzing this output, you can determine if the BGP route churn is contributing to the routing
```

```
protocol flap.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface 200.0.0.1 1 Full/ - 00:00:32 200.0.0.1
Ethernet3, Description: This command displays a list of OSPF
neighbors, including their state and dead time. By analyzing this
output, you can determine if the OSPF adjacency loss is contributing
to the routing protocol flap.

Command: show isis database, Expected Output: ISIS LSP Database: LSP
ID Sequence Checksum Age Holdtime Metric Next-Hop 1.1.1.1.1.1.1.1
0x0000000a 0x12345678 10 30 0 200.0.0.1, Description: This command
displays the ISIS LSP database, including the sequence number,
checksum, age, and next-hop information. By analyzing this output, you
can determine if the ISIS LSP flooding is contributing to the routing
protocol flap.
```

Resolution Steps

Step 1: Step: Verify the RIB and FIB consistency, Command: show ip route, Description: This command displays the IP routing table, which can be used to verify the consistency between the RIB and FIB. If the FIB is not consistent with the RIB, it may indicate a routing protocol flap.

Step 2: Step: Clear the BGP RIB and FIB, Command: clear bgp *, Description: This command clears the BGP RIB and FIB, which can help to resolve the BGP route churn contributing to the routing protocol flap.

Step 3: Step: Reset the OSPF neighbor, Command: clear ospf neighbor 200.0.0.1, Description: This command resets the OSPF neighbor connection, which can help to resolve the OSPF adjacency loss contributing to the routing protocol flap.

Step 4: Step: Reset the ISIS LSP, Command: clear isis database 1.1.1.1.1.1.1.1, Description: This command resets the ISIS LSP, which can help to resolve the ISIS LSP flooding contributing to the routing protocol flap.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP table version is 2,
main routing table version is 2 Neighbor V AS MsgRcvd MsgSent TblVer
InQ OutQ Up/Down State | bgp.200.0.0.1 4 200 10 10 2 0 0 00:02:11 4w4d
4 10/10, Description: This command displays a summary of the BGP
neighbor connections, including the number of messages received and
sent, as well as the table version. By analyzing this output, you can
determine if the BGP route churn has been resolved.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface 200.0.0.1 1 Full/ - 00:00:32 200.0.0.1
Ethernet3, Description: This command displays a list of OSPF
neighbors, including their state and dead time. By analyzing this
output, you can determine if the OSPF adjacency loss has been
resolved.

Command: show isis database, Expected Output: ISIS LSP Database: LSP
ID Sequence Checksum Age Holdtime Metric Next-Hop 1.1.1.1.1.1.1.1
0x0000000a 0x12345678 10 30 0 200.0.0.1, Description: This command
displays the ISIS LSP database, including the sequence number,
checksum, age, and next-hop information. By analyzing this output, you
can determine if the ISIS LSP flooding has been resolved.
```

Roll-back Procedures

- Step: Restore the previous RIB and FIB configuration, Command: copy running-config startup-config, Description: This command restores the previous RIB and FIB configuration from the startup-config file, which can help to roll back the changes made during the resolution process.

- Step: Clear the BGP RIB and FIB, Command: clear bgp *, Description: This command clears the BGP RIB and FIB, which can help to roll back the changes made during the resolution process.

- Step: Reset the OSPF neighbor, Command: clear ospf neighbor 200.0.0.1, Description: This command resets the OSPF neighbor connection, which can help to roll back the changes made during the resolution process.

- Step: Reset the ISIS LSP, Command: clear isis database 1.1.1.1.1.1.1.1, Description: This command resets the ISIS LSP, which can help to roll back the changes made during the resolution process.